

AutoTraQ — Railway Deployment Guide

Architecture

Railway runs two services from this monorepo:

Service	Directory	Port
Backend (Express API)	backend/	\$PORT (Railway-assigned)
Frontend (Vite/React)	frontend/	\$PORT (Railway-assigned)
MySQL	Railway add-on	3306

Step 1 — Create Railway Project

1. Go to railway.app and create a **New Project**.
2. Connect your GitHub repo.

Step 2 — Provision MySQL

1. In the project dashboard, click **+ New** → **Database** → **MySQL**.
2. Railway auto-creates a `MYSQL_URL` variable. You'll reference this below.

Step 3 — Deploy the Backend

1. Click **+ New** → **GitHub Repo** → select this repo.
2. Set **Root Directory** to `backend`.
3. Railway will detect `backend/railway.json` automatically.

Environment Variables

Variable	Value
<code>DATABASE_URL</code>	Copy from the MySQL service's <code>MYSQL_URL</code> (or use Railway's variable reference: <code>\${{MySQL.MYSQL_URL}}</code>)

Variable	Value
JWT_SECRET	Generate a random string (<code>openssl rand -hex 32</code>)
FRONTEND_URL	Your frontend URL, e.g. <code>https://cs490unco.org</code>
ENABLE_STARTUP_SEED	Optional. Set to <code>true</code> only for a fresh demo database that needs seed accounts.
PORT	Leave unset — Railway injects this automatically
NODE_ENV	<code>production</code>

What Happens on Deploy

Per `railway.json`: - **Build**: `npm install → prisma generate → tsc` - **Start**: `prisma db push → node dist/index.js`

This auto-applies schema changes to the database on every deploy.

Step 4 — Deploy the Frontend

1. Click + **New** → **GitHub Repo** → select this repo again.
2. Set **Root Directory** to `frontend`.

Environment Variables

Variable	Value
VITE_API_URL	Your backend's Railway URL, e.g. <code>https://autotraq-backend-production.up.railway.app</code>

⚠ VITE_ vars are baked in at **build time**. If you change the backend URL, redeploy the frontend.

What Happens on Deploy

Per `railway.json`: - **Build**: `npm install → tsc && vite build` - **Start**: `vite preview --host 0.0.0.0 --port $PORT`

Step 5 — Run Migrations & Seed

After the first successful backend deploy:

1. Open the backend service in Railway.
2. Go to the **Settings** tab → **Run Command** (or use Railway CLI).
3. Run one-off commands via Railway CLI:

```
# Install Railway CLI
npm i -g @railway/cli
railway login
railway link # select your project

# Run from backend directory
cd backend
railway run npx prisma db push
railway run npm run db:seed-admins
railway run npm run db:seed-vehicles
```

Or trigger seeds by temporarily changing the start command:

```
npx prisma db push --accept-data-loss && npm run db:seed-admins && npm run start
```

Then revert after the first deploy.

Step 6 — Generate Public URLs

1. For each service, go to **Settings** → **Networking** → **Generate Domain**.
 2. Note both URLs. Set the backend URL as `VITE_API_URL` on the frontend service.
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Default Admin Accounts

After seeding (`npm run db:seed-admins`):

Email	Password	Role
acordeiro@autotraq.app	autotraq2026	admin
team1@autotraq.app	autotraq2026	admin
team2@autotraq.app	autotraq2026	admin
team3@autotraq.app	autotraq2026	admin

Email	Password	Role
team4@autotraq.app	autotraq2026	admin

Change these passwords immediately after first login.

Troubleshooting

- **Build fails on Prisma:** Make sure `DATABASE_URL` is set before build. Prisma generate doesn't need the DB, but `db push` does.
- **Frontend can't reach backend:** Check `VITE_API_URL` is correct and the backend has CORS configured for the frontend domain.
- **MySQL connection refused:** Ensure `DATABASE_URL` uses the Railway internal URL (not public) for service-to-service communication.